

Program fundamental Lab report # 1



SUBMITTED BY:

ZULQARNAIN HASSAN

SUBMITTED TO:

SIR AAQIB ADEEL

System ID:	NUML-S22-15676
Faculty:	Engineering and computer science
Department:	Software engineering
Section:	BSSE-1-A-afternoon

First week

Date 17-02-2022

Topics

- **Header file**
- **Pre-processor directive**
- **Cout**
- **Main function**
- **endl**

Header file:

C++ contains many header files. Header files contain information that is required by the program in which these are used.

For example

iostream.h, conio.h, cmath etc.

Syntax

```
#include<iostream>
```

Pre-processor directive:

- It is used to include header files at the beginning of the program.
- It is a code for the compiler to include a header file.

For example

```
#include etc.
```

The syntax of the preprocessor directive to include a header file in a program is.

```
#include<name of header file>
```

It begins with a #sign, followed by the word include, and then the name of the header file is written within angled brackets.

The main function:

- Every program must contain the main function.
- The execution of the program begins from the main function.

Syntax

Main()

{

Body of program

}

The cout statement:

The cout statement is used to output text or values on the screen. The keyword cout insertion operator on its right side is two less than signs (<<).

Syntax

Cout<<" hello world";

Endl:

Endl is used to insert a new line character and flush the stream.

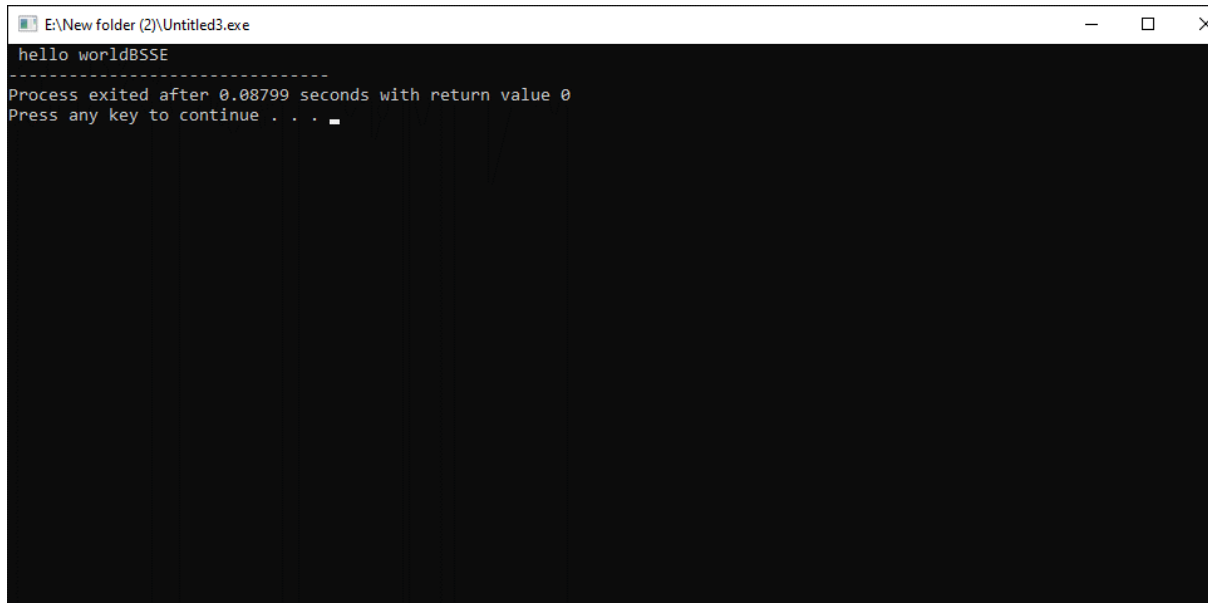
**Write a program that To print 'hello world and BSSE without
using endl**

```
#include<iostream>

using namespace std;

main()
{
cout<<" hello world";
cout<<"BSSE";
}
```

OUTPUT:-



Write a program that To print 'hello world and BSSE using endl

```
#include<iostream>

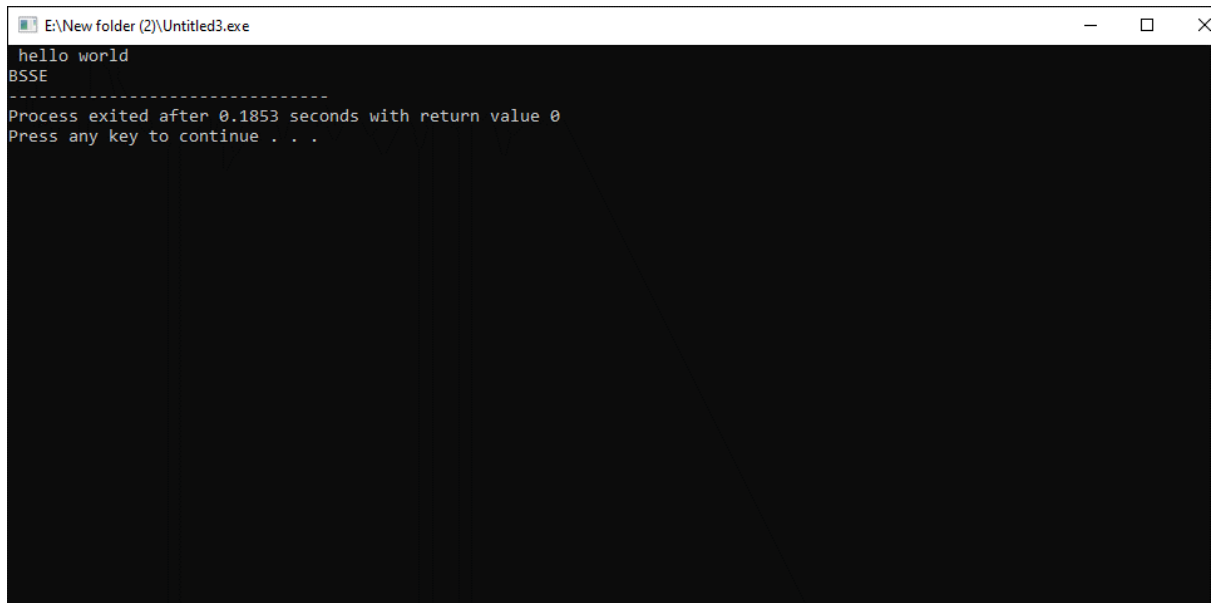
using namespace std;

main()
{
cout<<" hello world"<<endl;

cout<<"BSSE";

}
```

OUTPUT:-

A screenshot of a Windows command prompt window. The title bar shows the file path "E:\New folder (2)\Untitled3.exe". The window has standard Windows window controls (minimize, maximize, close). The command prompt displays the following text: "hello world", "BSSE", a line of dashes "-----", "Process exited after 0.1853 seconds with return value 0", and "Press any key to continue . . .". The rest of the window is black.

```
E:\New folder (2)\Untitled3.exe
hello world
BSSE
-----
Process exited after 0.1853 seconds with return value 0
Press any key to continue . . .
```

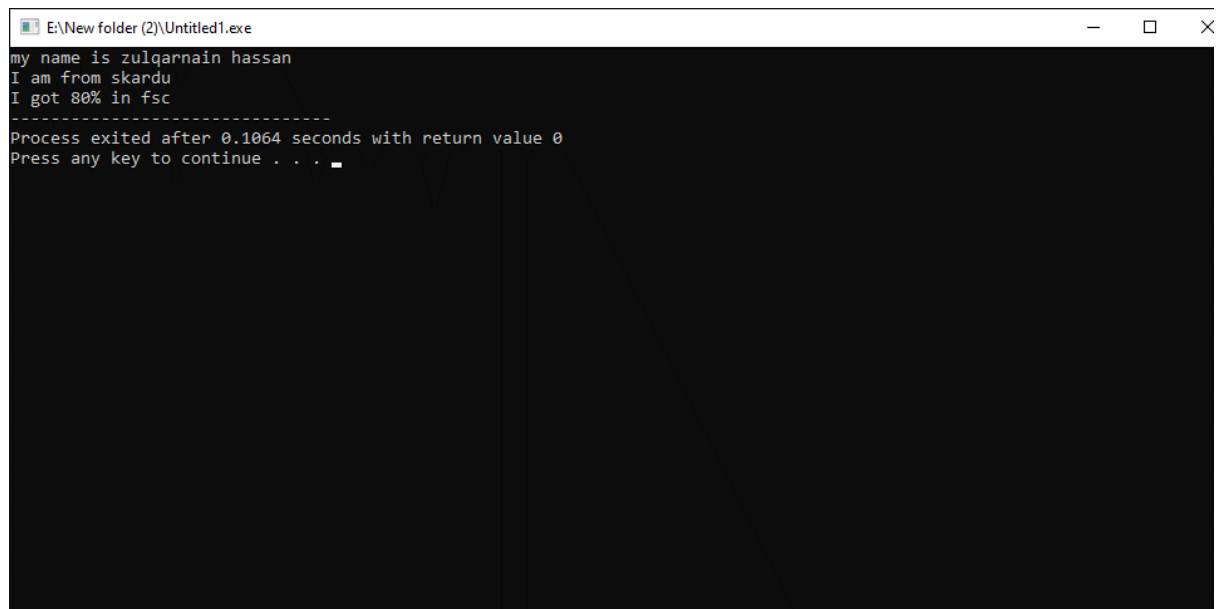
Write a program that print your Introduction.

```
#include<iostream>

using namespace std;

main()
{
    cout<<"my name is zulqarnain hassan"<<endl;
    cout<<"I am from skardu"<<endl;
    cout<<"I got 80% in fsc";
}
```

OUTPUT:-



A screenshot of a Windows command prompt window. The title bar at the top reads "E:\New folder (2)\Untitled1.exe" and includes standard minimize, maximize, and close buttons. The command prompt area has a black background with white text. The output of the program is as follows:

```
my name is zulqarnain hassan
I am from skardu
I got 80% in fsc
-----
Process exited after 0.1064 seconds with return value 0
Press any key to continue . . .
```

The cursor is positioned at the end of the "Press any key to continue . . ." line.

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Section:	BSSE-1-A-afternoon

second week

Date 22-02-2022

Topics

- Assignment operator
- Arithmetic operator
- \n and \t
- Int, float, double, string, char, bool

Assignment operator:

The assignment operator is used to assign values to variables.

Assignment operator: “=”.

Variable=expression.

For example

- A=b+c.
- B=123.

Arithmetic operator:

The arithmetic operator is used to perform arithmetic operations on variables and data.

Arithmetic operator:

+ addition

– subtraction

* Multiplication

/ Division

% Modulo operator

For example

A=34+g

f=a-b

B=3*6

c=a+b/4

\n and \t:

- \n is used to shifting to a new line.
- \t is used to give a tab space.

Int, float, double, string, char, bool.

- Int is used to store integers.
- Float is used to store decimal numbers with single precision.
- Double is used to store decimal numbers with double precision.
- A string is used to store a group of characters.
- Char is used to storing a character.
- Bool refers to a system of logical thought that is used to create true/false statements.

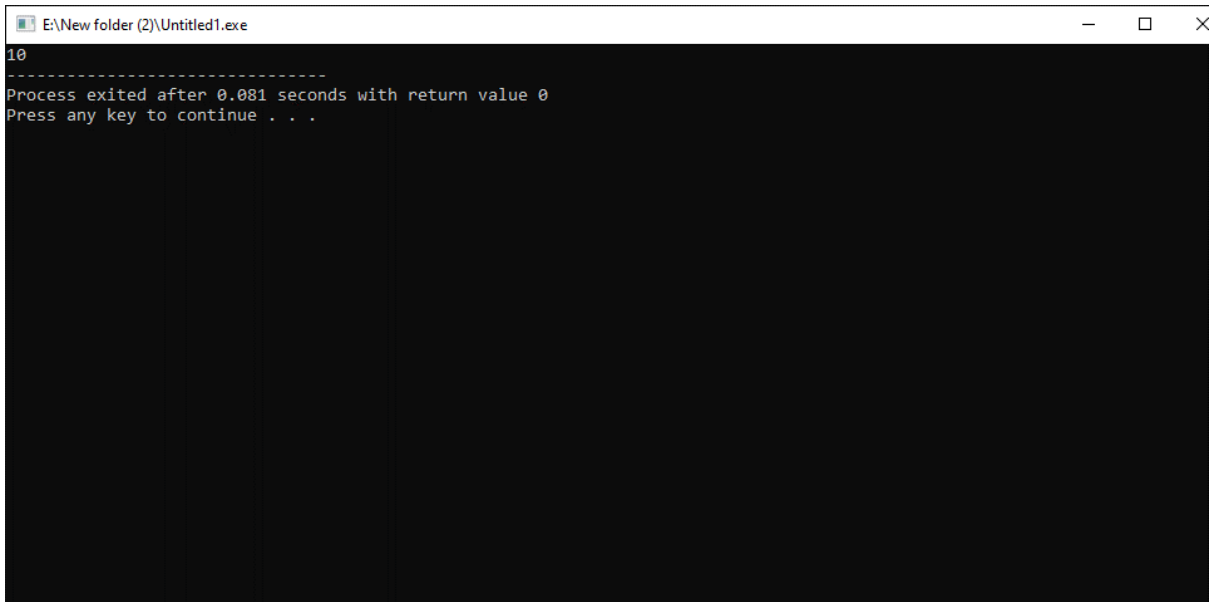
Write a program that assign an integer number in a variable
and print it

```
#include<iostream>

using namespace std;

main()
{
    int a;
    a=10
    cout<<a;
}
```

OUTPUT:-



```
E:\New folder (2)\Untitled1.exe
10
-----
Process exited after 0.081 seconds with return value 0
Press any key to continue . . .
```

Write a program that assign two decimal number in a variable
and print it

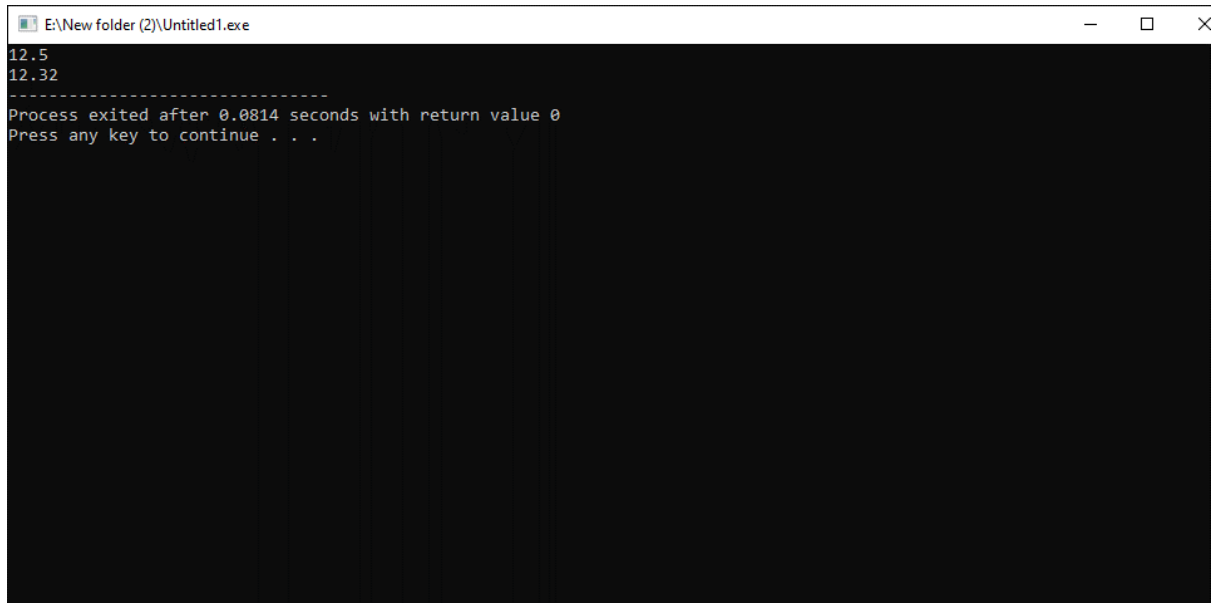
```
#include<iostream>

Using namespace std;

Main()
{
    Float a,b;
    a=12.5;
    b=12.32;
```

```
cout<<a<<b;  
  
}
```

OUTPUT:-



```
E:\New folder (2)\Untitled1.exe  
12.5  
12.32  
-----  
Process exited after 0.0814 seconds with return value 0  
Press any key to continue . . .
```

Write a program that assigns decimal numbers in variables and print them

```
#include<iostream>  
  
using namespace std;  
  
main()  
{  
double a=21.32451,b=23.4565,c=3.456223;  
float d=32.4563,e=32.445;  
cout<<a<<endl;  
cout<<b<<endl;  
cout<<c<<endl;  
cout<<d<<endl;  
cout<<e<<endl;  
}
```

OUTPUT:-

```
E:\New folder (2)\Untitled3.exe
21.3245
23.4565
3.45622
32.4563
32.445
-----
Process exited after 0.2947 seconds with return value 0
Press any key to continue . . .
```

Write a program that assigns a character in a variable and print it

```
4  #include<iostream>
    using namespace std;
    main()
    {
        char a;
        a='g';
        cout<<a;
    }OUTPUT:
```

```
E:\New folder (2)\Untitled1.exe
g
-----
Process exited after 0.07229 seconds with return value 0
Press any key to continue . . .
```

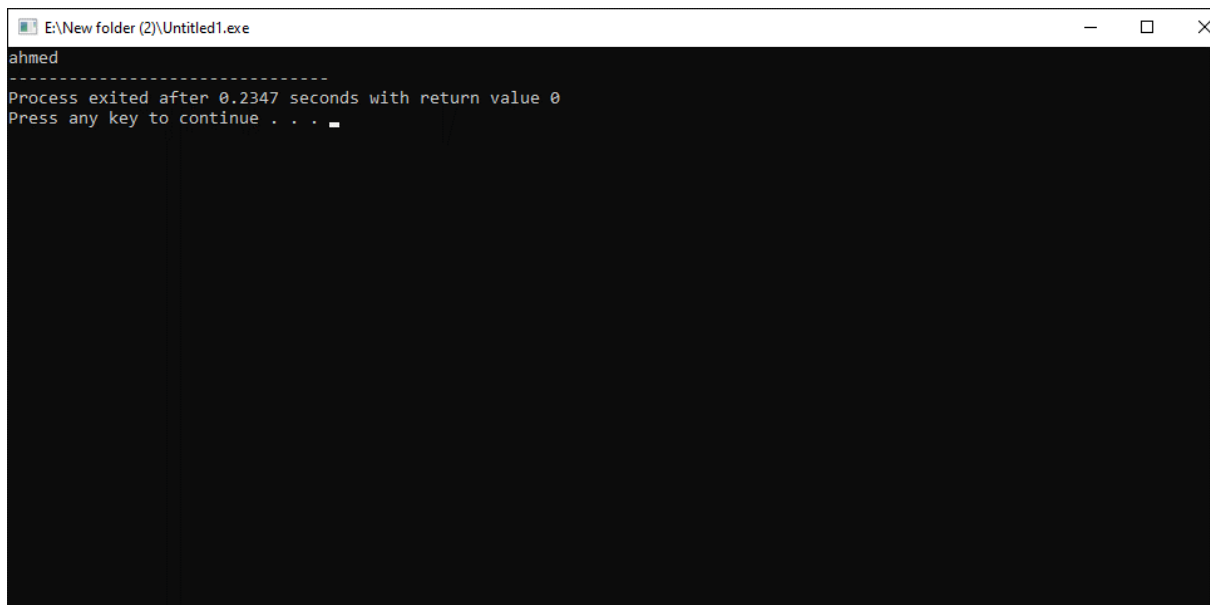
Write a program that assigns a string in a variable and print it

```
#include<iostream>

using namespace std;

main()
{
    string a;
    a="Ahmed";
    cout<<a;
}
```

OUTPUT:-

A screenshot of a Windows command prompt window titled "E:\New folder (2)\Untitled1.exe". The window has a black background with white text. The output of the program is displayed as follows: "ahmed" followed by a dashed line separator. Below the separator, it says "Process exited after 0.2347 seconds with return value 0" and "Press any key to continue . . .". A cursor is visible at the end of the "Press any key to continue" line.

```
E:\New folder (2)\Untitled1.exe
ahmed
-----
Process exited after 0.2347 seconds with return value 0
Press any key to continue . . .
```

Write a program that print “ahmad call ali” using string

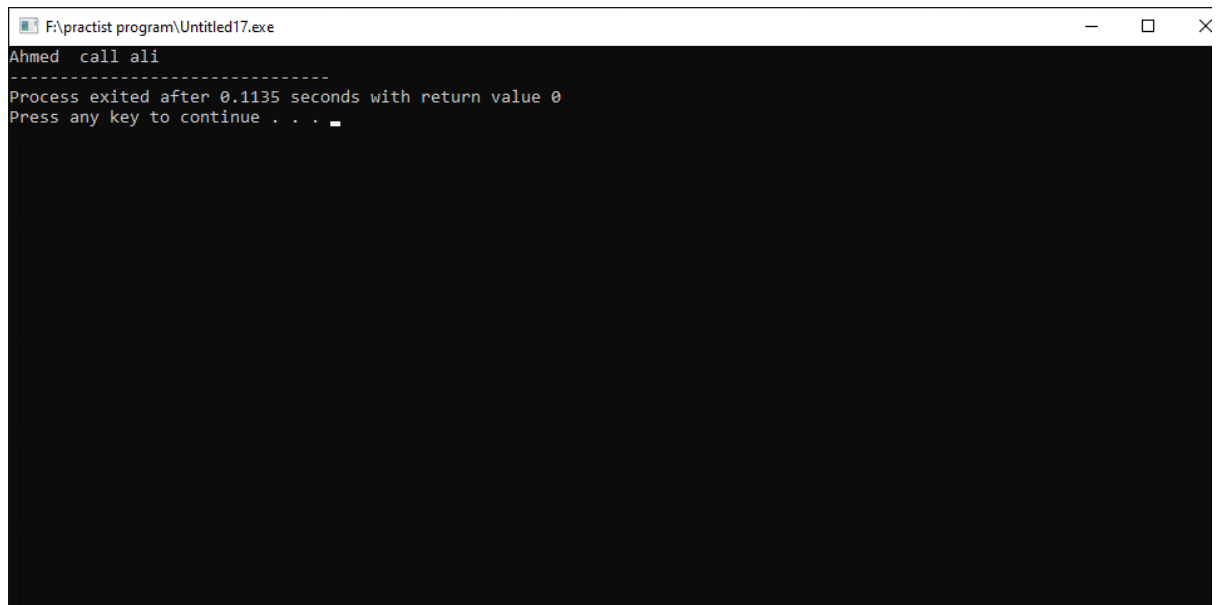
```
#include<iostream>

using namespace std;

main()
{
    string a,b;
    a="Ahmed";
    b="ali";

    cout<<a<<" call "<<b;
```

```
}
```

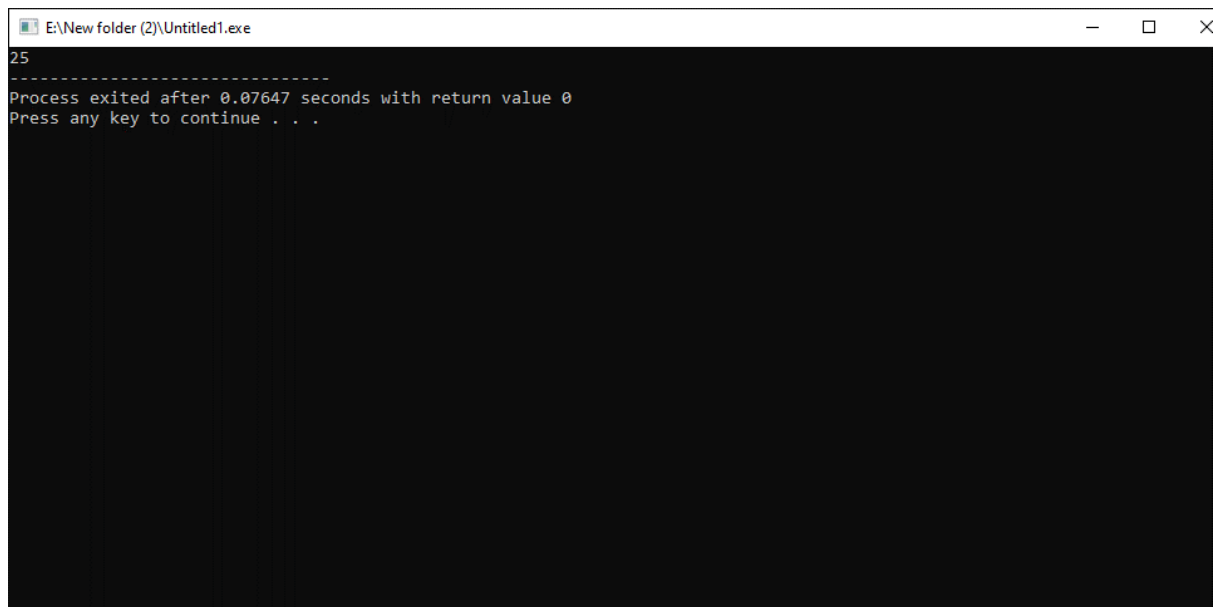


```
F:\practist program\Untitled17.exe
Ahmed call ali
-----
Process exited after 0.1135 seconds with return value 0
Press any key to continue . . .
```

**Write a program that assign two numbers in variables and
print thir sum**

```
6  #include<iostream>
using namespace std;
main()
{
    int a,b,sum;
    a=12;
    b=13;
    sum=a+b;
    cout<<sum;
}
```

OUTPUT:-



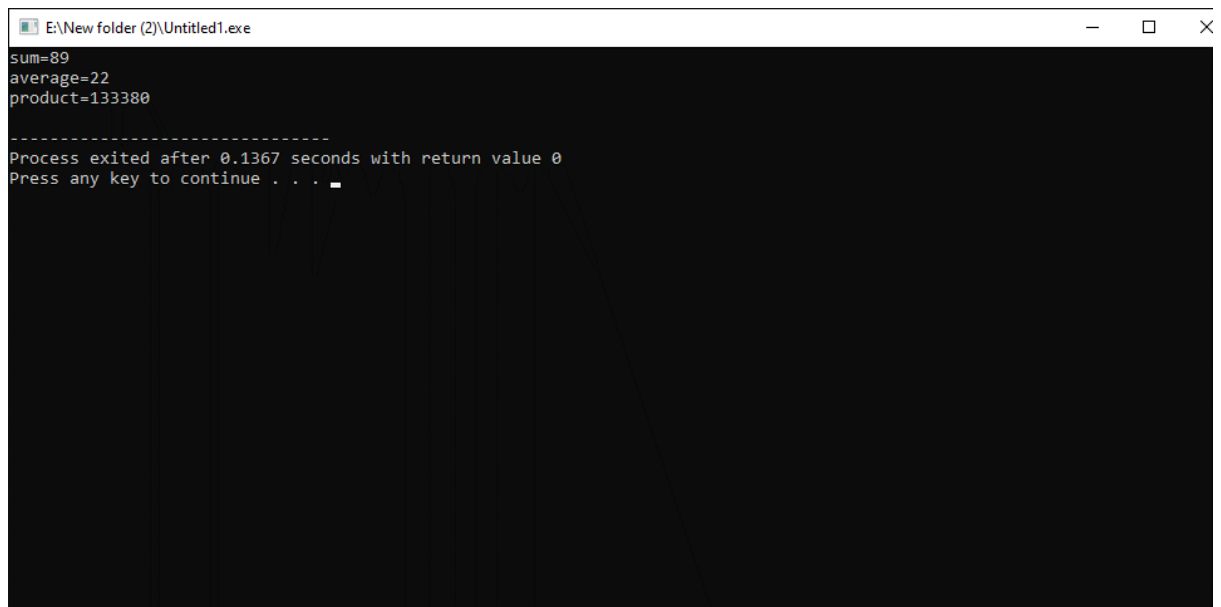
```
E:\New folder (2)\Untitled1.exe
25
-----
Process exited after 0.07647 seconds with return value 0
Press any key to continue . . .
```

Write a program that assigns four numbers in variables and prints their sum, average, and product

```
#include<iostream>
#include<cmath>
using namespace std;
main()
{
    int a,b,c,d,sum,average,product;
    a=12;
    b=13;
    c=45;
    d=19;
    sum=a+b+c+d;
    average=sum/4;
    product=a*b*c*d;
    cout<<"sum="<<sum<<"\n";
    cout<<"average="<<average<<"\n";
    cout<<"product="<<product<<"\n";

}
```

OUTPUT:-



The screenshot shows a Windows command prompt window titled "E:\New folder (2)\Untitled1.exe". The output of the program is as follows:

```
sum=89
average=22
product=133380
-----
Process exited after 0.1367 seconds with return value 0
Press any key to continue . . .
```

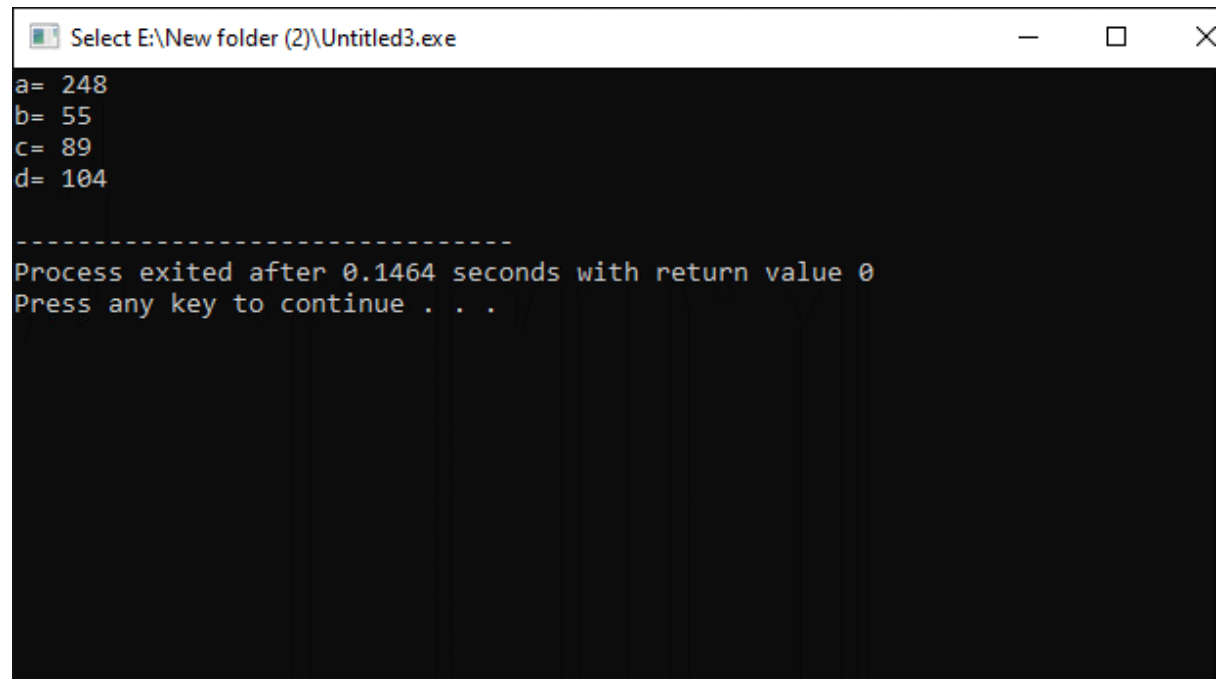
Write a program that performs the following operations

If a=12

b=a+43,c=b+34,d=a+c+3,a=b+c+d

```
#include<iostream>
#include<cmath>
using namespace std;
main()
{
    Int a,b,,c,d;
    a=12;
    b=a+43;
    c=b+34;
    d=a+c+3;
    a=b+c+d;
    cout<<"a= "<<a\n;
    cout<<"b= "<<b;
    cout<<"c= "<<c;
    cout<<"d= "<<d;
}
```


OUTPUT:-



The screenshot shows a Windows command prompt window with a title bar that reads "Select E:\New folder (2)\Untitled3.exe". The window has standard minimize, maximize, and close buttons. The command prompt displays the following text:

```
a= 248  
b= 55  
c= 89  
d= 104  
  
-----  
Process exited after 0.1464 seconds with return value 0  
Press any key to continue . . .
```

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Section:	BSSE-1-A-afternoon

third week

Date 28-02-2022

Topics in the third week

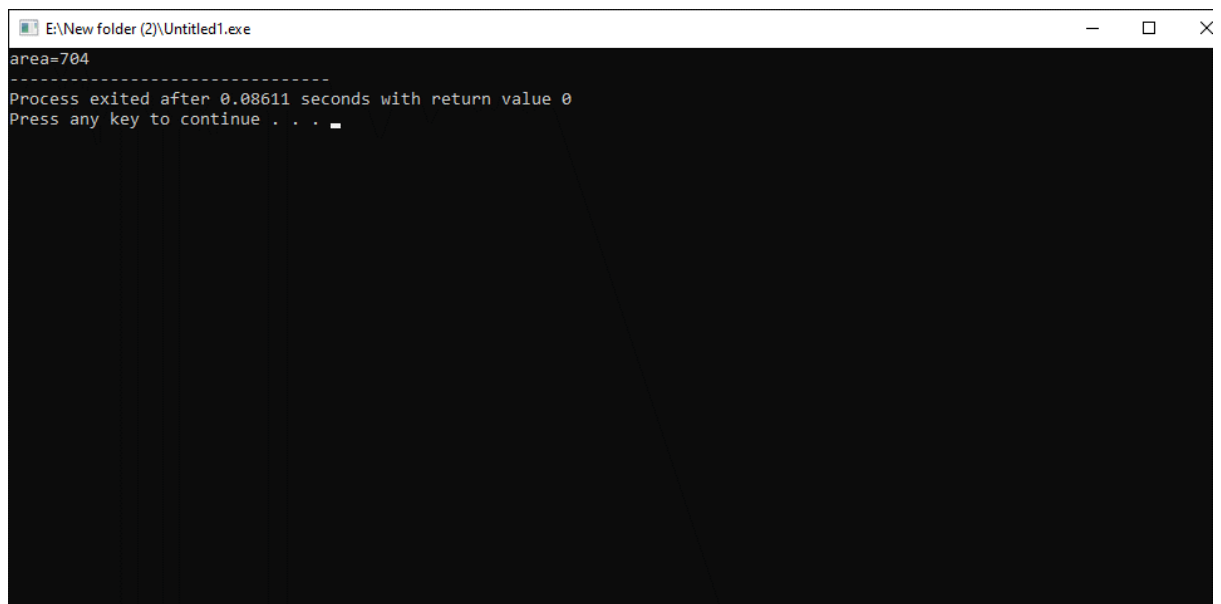
- 1 Calculate the area of the rectangle if length and width are given.
- 2 Calculate the area of a triangle if perpendicular and base are given.
- 3 Calculate the area and circumference of a circle if the radius is given.
- 4 Calculate sum, average, and percentage of a student's marks of six subjects
- 5 Convert rupee into the dollar, euro, and pound.
- 6 Find hypotenuse if altitude and base are given.
- 7 Calculate the perimeter of a rectangle.
- 8 Calculate the third angle if two angles are given.

Program NO 1

Calculate the area of the rectangle if length and width are given.

```
#include<iostream>
#include<cmath>
using namespace std;
main()
{
    int len,wid,area;
    len=32;
    wid=22;
    area=len*wid;
    cout<<"area="<<area;
}
```

OUTPUT:-



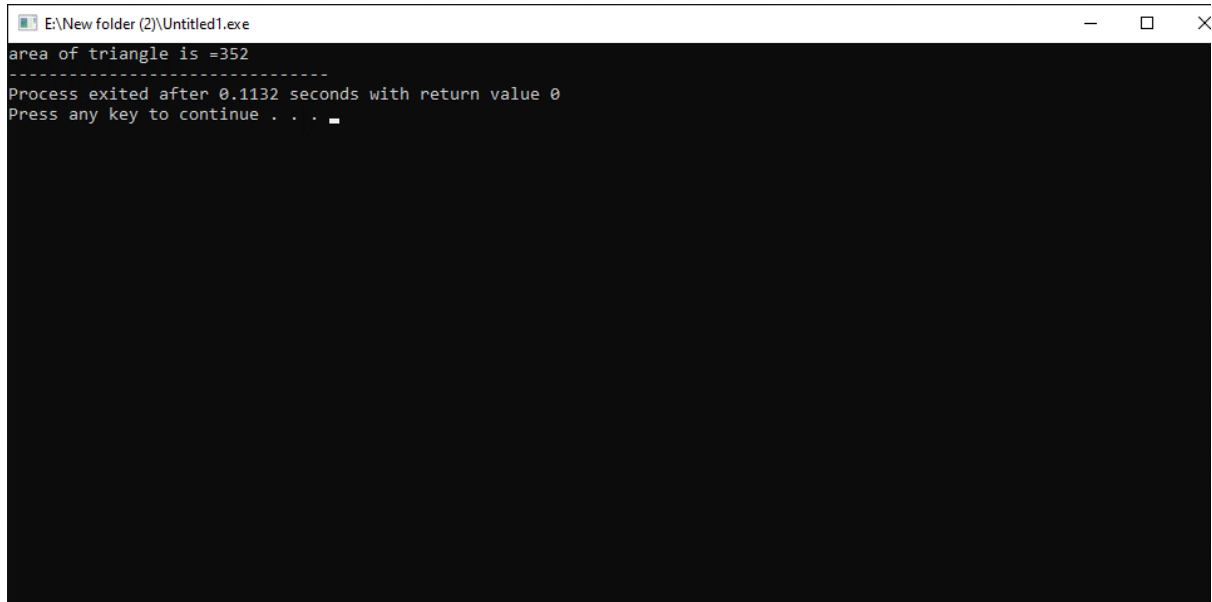
```
E:\New folder (2)\Untitled1.exe
area=704
-----
Process exited after 0.08611 seconds with return value 0
Press any key to continue . . .
```

Program NO 2

calculate area of triangle if perpendicular and base are given.

```
#include<iostream>
#include<cmath>
using namespace std;
main()
{
    int base,perp,area;
    base=32;
    perp=22;
    area=base*perp;
    cout<<"area of triangle is ="<<area;
}
```

OUTPUT:-

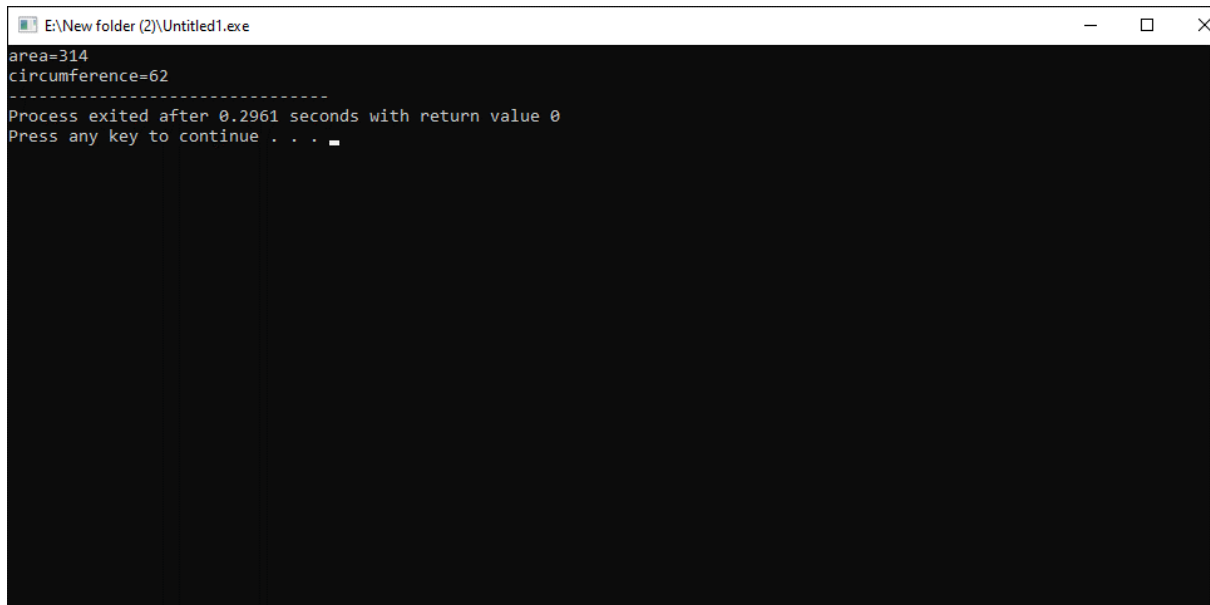


Program NO 3

Calculate the area and circumference of a circle if the radius is given.

```
#include<iostream>
#include<cmath>
using namespace std;
main()
{
    int radius=10,area,circumference;
    area=radius*radius*3.1415;
    circumference=2*3.1415*radius;
    cout<<"area="<<area<<endl<<"circumference="<<circumference;
}
```

OUTPUT:-

A screenshot of a Windows command prompt window. The title bar shows the file path "E:\New folder (2)\Untitled1.exe". The command prompt displays the output of a program: "area=314", "circumference=62", followed by a dashed line "-----". Below this, it says "Process exited after 0.2961 seconds with return value 0" and "Press any key to continue . . .". The cursor is positioned after the last line of text.

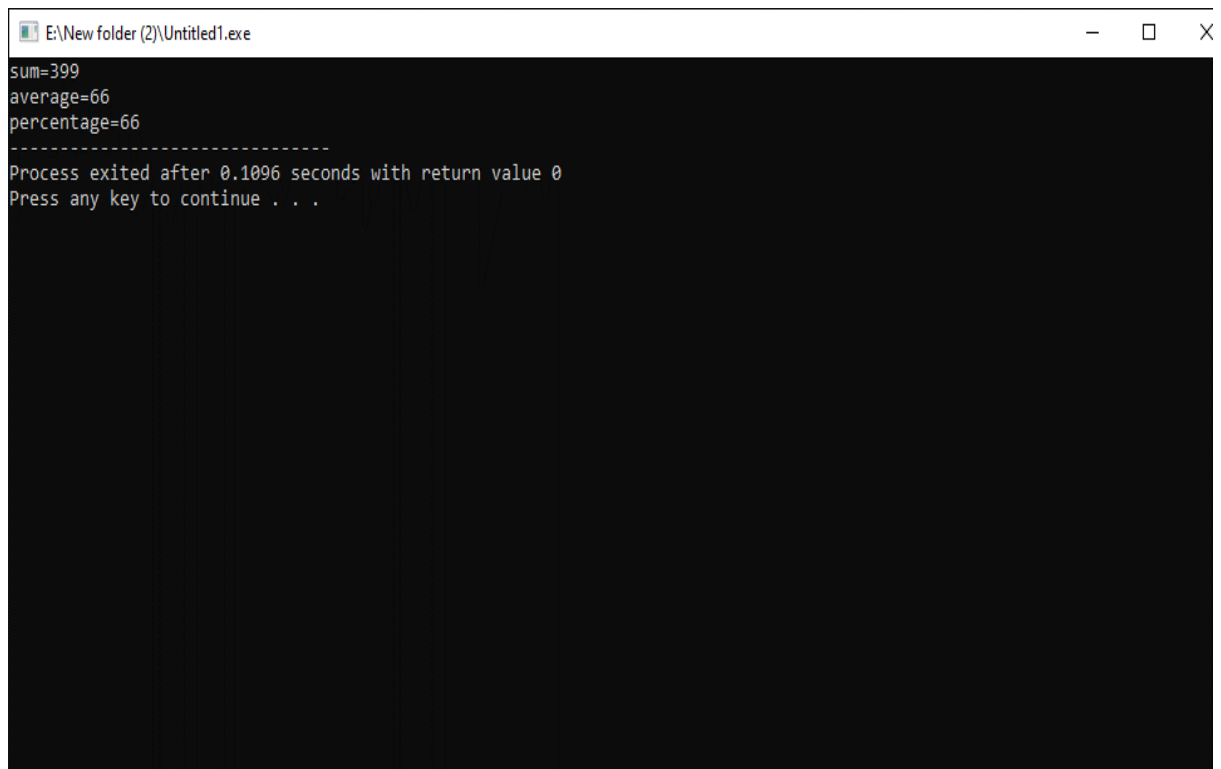
```
E:\New folder (2)\Untitled1.exe
area=314
circumference=62
-----
Process exited after 0.2961 seconds with return value 0
Press any key to continue . . .
```

Program NO 4

Calculate sum, average and percentage of a student marks of six subjects.

```
#include<iostream>
#include<cmath>
using namespace std;
main()
{
    int total=600,sub1,sub2,sub3,sub4,sub5,sub6,sum,average,percentage;
    sub1=64;
    sub2=79;
    sub3=34;
    sub4=67;
    sub5=90;
    sub6=65;
    sum=sub1+sub2+sub3+sub4+sub5+sub6;
    average=sum/6;
    percentage=100*sum/total;
    cout<<"sum="<<sum<<endl<<"average="<<average<<endl<<"percentag
e="<<percentage;
}
```

OUTPUT:-

A screenshot of a Windows command prompt window. The title bar shows the file path "E:\New folder (2)\Untitled1.exe". The window has standard Windows window controls (minimize, maximize, close). The output text is as follows:

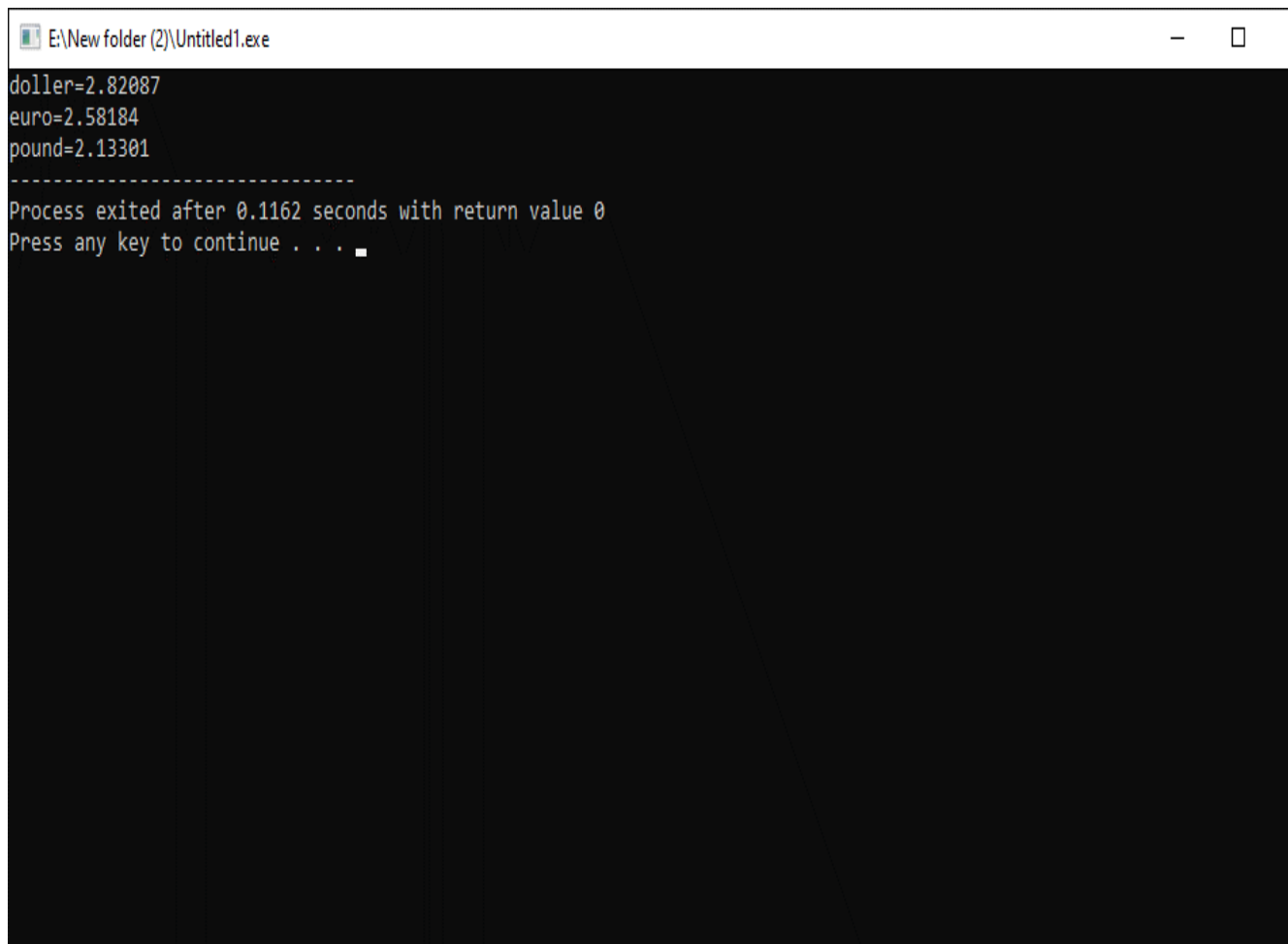
```
sum=399
average=66
percentage=66
-----
Process exited after 0.1096 seconds with return value 0
Press any key to continue . . .
```

Program NO 5

Convert rupee into doller, euro and pound

```
#include<iostream>
#include<cmath>
using namespace std;
main()
{
    double rupee,doller,euro,pound;
    rupee=500;
    doller=500/177.25;
    euro=500/193.66;
    pound=500/234.41;
    cout<<"doller="<<doller<<endl<<"euro="<<euro<<endl<<"pound="<<po
und;
}
```

OUTPUT:-

A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\New folder (2)\Untitled1.exe'. The command prompt displays the following text: 'doller=2.82087', 'euro=2.58184', 'pound=2.13301', followed by a line of dashes '-----'. Below the dashes, it says 'Process exited after 0.1162 seconds with return value 0' and 'Press any key to continue . . .'.

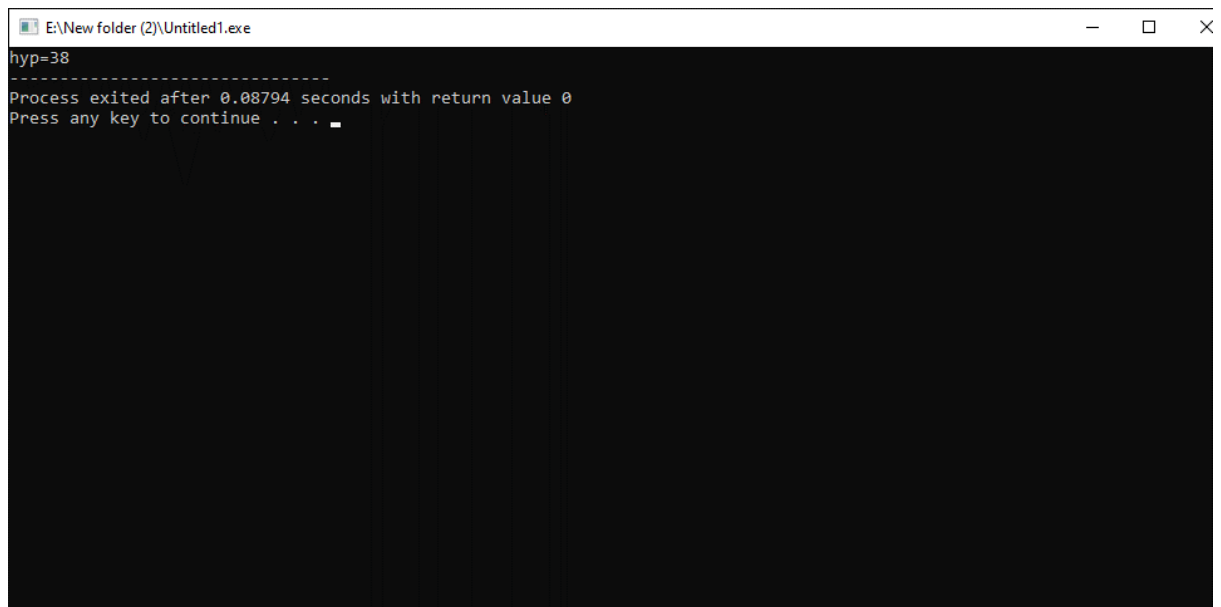
```
E:\New folder (2)\Untitled1.exe
doller=2.82087
euro=2.58184
pound=2.13301
-----
Process exited after 0.1162 seconds with return value 0
Press any key to continue . . .
```

Program NO 6

Find hypotenuse if altitude and base are given.

```
#include<iostream>
#include<cmath>
using namespace std;
main()
{
    int alt,base,hyp;
    alt=32;
    base=22;
    hyp=sqrt((alt*alt)+(base*base));
    cout<<"hyp="<<hyp;
}
```

OUTPUT:-

A screenshot of a Windows command prompt window. The title bar shows the file path "E:\New folder (2)\Untitled1.exe". The window has standard Windows window controls (minimize, maximize, close). The command prompt shows the output of a program: "hyp=38", followed by a dashed line separator "-----", then "Process exited after 0.08794 seconds with return value 0", and finally "Press any key to continue . . .". A cursor is visible at the end of the last line.

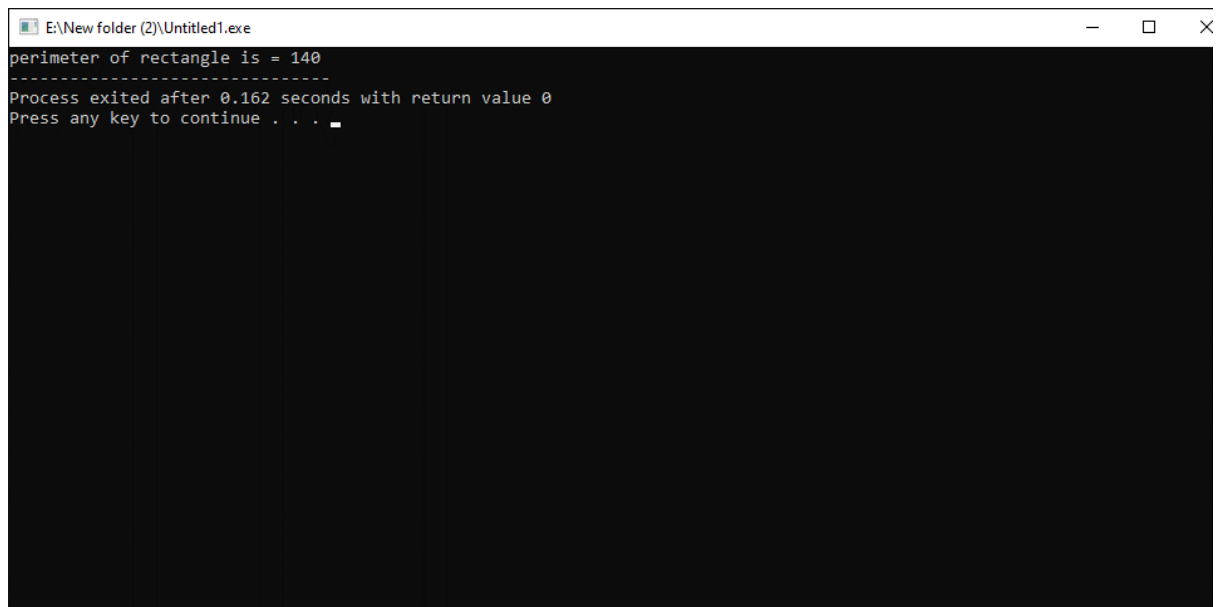
```
E:\New folder (2)\Untitled1.exe
hyp=38
-----
Process exited after 0.08794 seconds with return value 0
Press any key to continue . . .
```

Program NO 7

Calculate the perimeter of a rectangle.

```
#include<iostream>
using namespace std;
main()
{
int len,wid,perimeter;
len=40;
wid=30;
perimeter=2*(len+wid);
cout<<"perimeter of rectangle is = "<<perimeter;
}
```


OUTPUT:-



```
E:\New folder (2)\Untitled1.exe
perimeter of rectangle is = 140
-----
Process exited after 0.162 seconds with return value 0
Press any key to continue . . .
```

Program NO 8

Calculate the third angle of a triangle if two angles are given.

```
#include<iostream>
using namespace std;
main()
{
    int angle1,angle2,angle3;
    angle1=90;
    angle2=30;
    angle3=180-angle1-angle2;
    cout<<"angle3="<<angle3;
}
```

OUTPUT:-

angle3=60

Process exited after 0.158 seconds with return value 0

Press any key to continue . . .

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Section:	BSSE-1-A-afternoon

fourth week

Date 07-03-2022

- ❖ cin.
- ❖ Unary operator.
- ❖ Binary operator.
- ❖ Ternary operator.
- ❖ If statement.
- ❖ Multiple If statement.
- ❖ Nested if statement.
- ❖ If-else statement.
- ❖ Else-if statemen.

Cin:

The cin object is used to accept input from the user.

Unary operator:

The operator that works on a single operand is known as a unary operator.

Some examples of unary operators are.

A=b++

B=-a

--x

Binary operator:

The operator that works on two operators is known as the binary operator.

Some examples of binary operators are.

A=b-c

V=a*b

V=b%e

Ternary operator:

The conditional operator (? :) is known as the ternary operator since it has three parts.

These three parts are a condition and two expressions. The condition is evaluated and based on its result one of the expressions is executed.

If statement:

The if statement is used to execute a block of statements based on a condition.

Syntax

If(condition)

{

Block of statements

}

Multiple IF statements:

If we using more than one if statement in a function called multiple if statement.

Syntax

If(condition)

{

Block of statements

}

If(condition)

{

Block of statements

}.....

Nested-If statement

if statement inside another if statement is called nested if statement.

Syntax

If(condition)

{

If(condition)

{

body

}

}

If-else statement:

If else allows making decision between two courses of action based on a condition.

The following is the general form of if-else statement.

If(condition)

{

Block of statement.

}

else

{

Block of statement.

}

else –If statement:

The else if statement is used in situation where a decision is to be made from several alternatives based on various conditions.

If(condition)

{

Block of statement 1

}

Else if(condition)

{

Block of statement 2

}

.

.

.

Else

{

Block of statement- n

}

Write a program that get the length and width of a rectangle and print its area

```
#include<iostream>

using namespace std;

main()
{
    int len,wid,area;

    cout<<"enter length ";

    cin>>len;

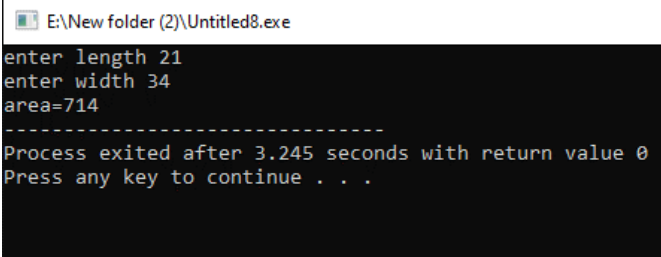
    cout<<"enter width ";

    cin>>wid;

    area=len*wid;

    cout<<"area="<<area;

}
```



Write a program that get two angles of a triangle and print its third angle

```
#include<iostream>

using namespace std;

main()
{
    int a1,a2,a3;

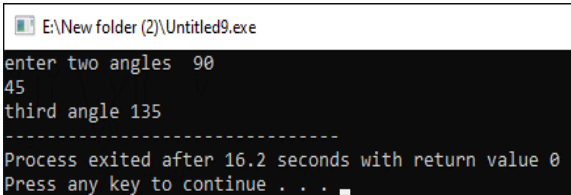
    cout<<"enter two angles ";

    cin>>a1>>a2;

    a3=180-a1+a2;

    cout<<"third angle "<<a3;

}
```



Write a program that get radius from the user and print its circumference

```
#include<iostream>

using namespace std;

main()
{
```

```

float r,c;

cout<<"enter radius ";

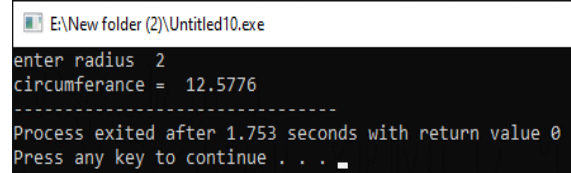
cin>>r;

c=2*3.1444*r;

cout<<"circumference = "<<c;

}

```



```

E:\New folder (2)\Untitled10.exe
enter radius 2
circumference = 12.5776
-----
Process exited after 1.753 seconds with return value 0
Press any key to continue . . .

```

Write a program that get a number and print with an increment

```

#include<iostream>

using namespace std;

main()

{

    int a;

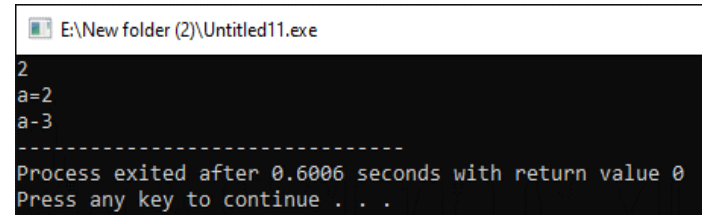
    cin>>a;

    cout<<"a="<<a++;

    cout<<"\na-"<<a;

}

```



```

E:\New folder (2)\Untitled11.exe
2
a=2
a-3
-----
Process exited after 0.6006 seconds with return value 0
Press any key to continue . . .

```

Write a program that get a number and print with a decrement

```

#include<iostream>

using namespace std;

main()

{

    int a;

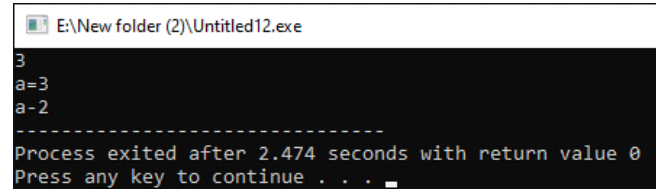
    cin>>a;

    cout<<"a="<<a--;

    cout<<"\na-"<<a;

}

```



```

E:\New folder (2)\Untitled12.exe
3
a=3
a-2
-----
Process exited after 2.474 seconds with return value 0
Press any key to continue . . .

```

Write a program that get a number and print with an increment

```

#include<iostream>

using namespace std;

main()

{

```



```

        int a,b;

        cin>>a;

        cout<<"a="<<--a;

        b=++a;

        cout<<"\na="<<b;

    }

```

```

E:\New folder (2)\Untitled13.exe
4
a=3
a=4
-----
Process exited after 0.9739 seconds with return value 0
Press any key to continue . . .

```

Write a program that get the age from user and print “you can have CNIC” if age is greater than 18.

```

#include<iostream>

using namespace std;

main()
{

    int age;

    cout<<"enter your age ";

    cin>>age;

    if(age>18)
    {

        cout<<"you can have CNIC";

    }

}

```

```

E:\New folder (2)\Untitled14.exe
enter your age 22
you can have CNIC
-----
Process exited after 1.641 seconds with return value 0
Press any key to continue . . .

```

Write a program that ask the user to guess a number and if number is equale to 18 then print “you guess the right number ”

```

#include<iostream>

using namespace std;

main()
{

    int num;

    cout<<"enter a number ";

    cin>>num;

    if(num==18)
    {

        cout<<"you guess the right number";

    }

}

```

```

E:\New folder (2)\Untitled14.exe
enter a number 18
you guess the right number
-----
Process exited after 4.044 seconds with return value 0
Press any key to continue . . .

```

```

    }
}

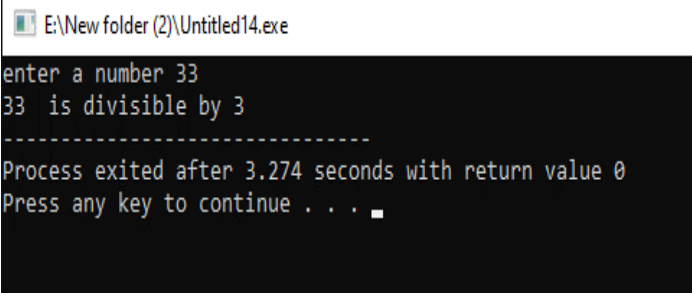
```

Write a program that get a number from user and if the number is divisible by 3 then print number is divisible by 3

```

#include<iostream>
using namespace std;
main()
{
    int num;
    cout<<"enter a number ";
    cin>>num;
    if(num%3==0)
    {
        cout<<num<<" is divisible by 3";
    }
}

```



```

E:\New folder (2)\Untitled14.exe
enter a number 33
33 is divisible by 3
-----
Process exited after 3.274 seconds with return value 0
Press any key to continue . . .

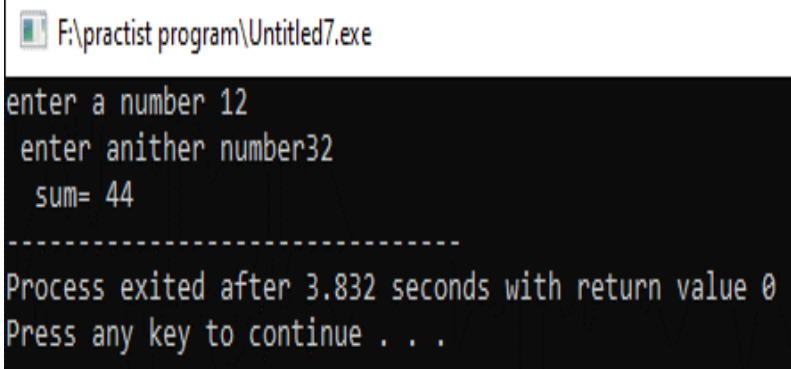
```

Write a program that read two numbers. If the first number is a positive number then it will print the sum of the two numbers.

```

#include<iostream>
using namespace std;
main()
{
    int n1,n2,sum;
    cout<<"enter a number ";
    cin>>n1;
    cout<<" enter anither number";
    cin>>n2
    if(n1%2==0)
    {
        Sum=n1+n2;
        cout <<" sum= "<<sum;
    }
}

```



```

F:\practist program\Untitled7.exe
enter a number 12
enter anither number32
sum= 44
-----
Process exited after 3.832 seconds with return value 0
Press any key to continue . . .

```

**Write a program that get a number from user and if the number is divisible by 5 then
print number is divisible by 5**

```
#include<iostream>

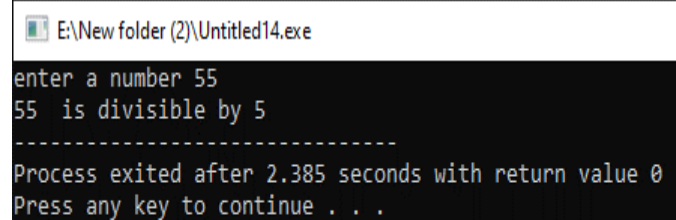
using namespace std;

main()
{
    int num;

    cout<<"enter a number ";

    cin>>num;

    if(num%5==0)
    {
        cout<<num<<" is divisible by 5";
    }
}
```



```
E:\New folder (2)\Untitled14.exe
enter a number 55
55 is divisible by 5
-----
Process exited after 2.385 seconds with return value 0
Press any key to continue . . .
```

Write a program that get a number from user and perform the following operation

b=a-2;

c=b*a*4+b;

d=c*a+12*d;

a=c*d-a*d+12;

e=d-d*a-a*b-b;

f=a+e-b*c;

a=a*f;

b=a+b+c+d+f;

```
#include<iostream>

using namespace std;

main()
{
    int a,b,c,d,e,f;

    cin>>a;
```

```

b=a-2;

c=b*a*4+b;

d=c*a+12*d;

a=c*d-a*d+12;

e=d-d*a-a*b-b;

f=a+e-b*c;

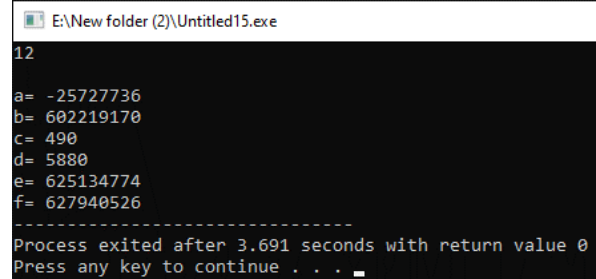
a=a*f;

b=a+b+c+d+f;

cout<<"\na= "<<a<<"\nb= "<<b<<"\nc= "<<c<<"\nd= "<<d<<"\ne= "<<e<<"\nf= "<<f;

}

```



Write a program that read the section from the user and prints the room number if the section is A then room 42 and if the section is B then room 43.

```

#include<iostream>

Using namespace std;

Main()
{
Char section;

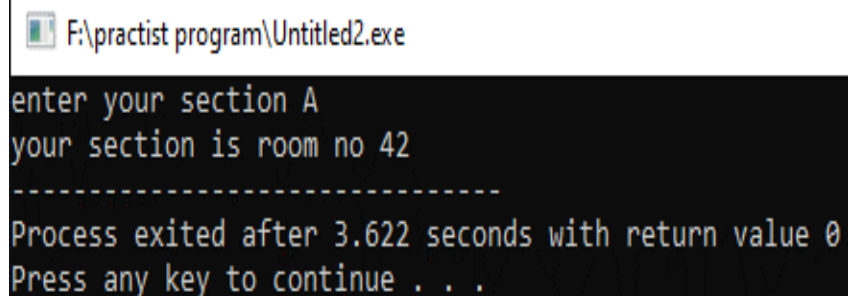
Cout<<"enter your section";

Cin>>section;

If(section==A)
{
Cout<<"your section is room no 42";
}

If(section==B)
{
Cout<<" your section is room no 43";
}
}

```



Write a program that gets three numbers from the user and print the greatest number

```

#include<iostream>

using namespace std;

main()

```

```

{
int a,b,c;

cout<<"enter three number";

cin>>a>>b>>c;

cout<<"a= "<<a<<"\n b= "<<b<<"\n c= "<<c<<endl;

if(a>b)
{
if(a>c)
{

cout<<a<<" is greatest";

}

}

if(b>a)
{
if(b>c)
{

cout<<b<<" is greatest";

}

}

if(c>a)
{
if(c>b)
{

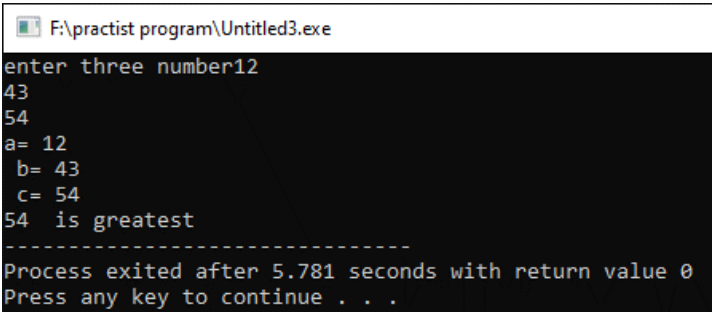
cout<<c<<" is greatest";

}

}

}
}

```



```

F:\practist program\Untitled3.exe
enter three number12
43
54
a= 12
b= 43
c= 54
54 is greatest
-----
Process exited after 5.781 seconds with return value 0
Press any key to continue . . .

```

Write a program that read a three-digit number from the user and prints its last digit.

```

#include<iostream>

using namespace std;

main()

```

```

{
int num,digit;

cout<<"enter three digit number";

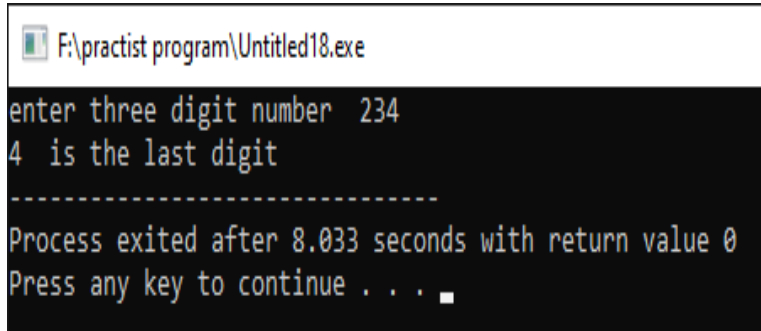
cin>>num;

digit=num%10;

cout<<digit<<" is the last digit";

}

```



```

F:\practist program\Untitled18.exe
enter three digit number 234
4 is the last digit
-----
Process exited after 8.033 seconds with return value 0
Press any key to continue . . .

```

Write a program that read a number between 1 and 10 and print as:

- **Number > 7 : print too high.**
- **umber < 7 :print too less.**
- **number =7 : you guess the write number**
 - **other wice invalide**

```

#include<iostream>

Using namespace std;

Main()

{
int num;

Cout<<"guess a number";

Cin>>num;

If(num<=10)

{

If(num>=0)

{

If(num<7)

{

cout<<" too low";

}

If(num>7)

{

Cout<<" too high";

}

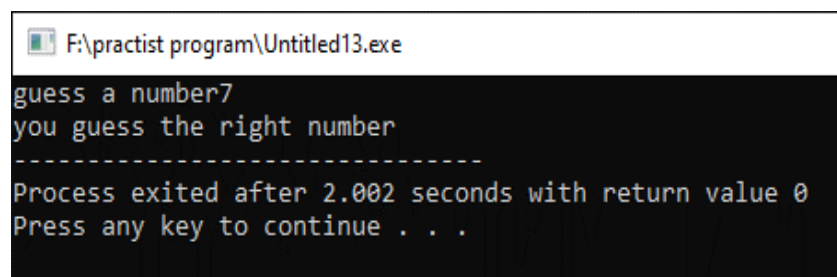
Else

{

Cout<<"you guess the right number";

}

```



```

F:\practist program\Untitled13.exe
guess a number7
you guess the right number
-----
Process exited after 2.002 seconds with return value 0
Press any key to continue . . .

```

```
}  
}  
}
```

Write a program that read a number and print as

- 1 is factor of n
- 2 is factor of n and so on up to 10.

```
#include<iostream>  
  
Using namespace std;  
  
Main()  
{  
    Int n;  
    Cout<<" enter a number";  
    Cin>>n;  
    If(n%1==0)  
    {  
        Cout<<"\n 1 is factor of " <<n;  
    }  
    If(n%2==0)  
    {  
        Cout<<"\n 2 is factor of " <<n;  
    } If(n%3==0)  
    {  
        Cout<<"\n 3 is factor of " <<n;  
    } If(n%4==0)  
    {  
        Cout<<"\n 4 is factor of " <<n;  
    } If(n%5==0)  
    {  
        Cout<<"\n 5 is factor of " <<n;  
    } If(n%6==0)  
    {  
        Cout<<"\n 6 is factor of " <<n;  
    } If(n%7==0)  
    {  
        Cout<<"\n 7 is factor of " <<n;
```

```

} If(n%8==0)
{
    Cout<<"\n 8 is fector of " <<n;
} If(n%9==0)
{
    Cout<<"\n 9 is fector of " <<n;
} If(n%10==0)
{
    Cout<<"\n 10 is fector of " <<n;
}
}
}

```

F:\practist program\Untitled14.exe

enter a number100

```

1 is fector of 100
2 is fector of 100
4 is fector of 100
5 is fector of 100
10 is fector of 100

```

 Process exited after 4.677 seconds with return value 0
 Press any key to continue . . .

Write a program that read your salary and print as:

If salary=50000 then tex=45%

If salary>50000 then tex=70%

If salary<50000 then tex=35%

```

#include<iostream>

using namespace std
main()
{
    int salary,tex;
    cout<<"enter your salary";
    cin>>salary;
    if(salary==50000)
    {
        tex=50000*45/100;
        cout<<"tex is= " <<tex;
        cout<<"\nnow your salary is = " <<salary-tex;
    }
    if(salary>50000)
    {
        tex=50000*70/100;
        cout<<"tex is= " <<tex;
    }
}

```



```

cout<<"\nnow your salary is = "<<salary-tex;

}

if(salary<50000)

{

tex=50000*35/100;

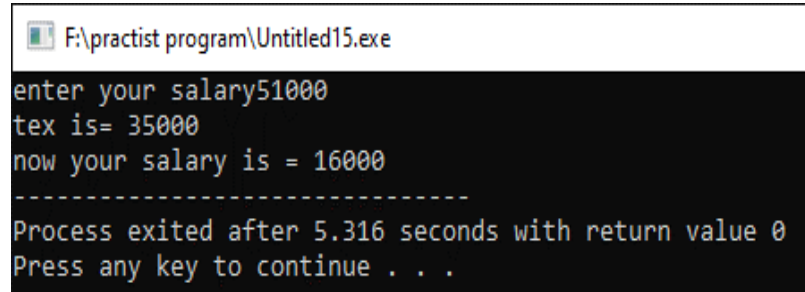
cout<<"tex is= "<<tex;

cout<<"\nnow your salary is = "<<salary-tex;

}

}

```



```

F:\practist program\Untitled15.exe
enter your salary51000
tex is= 35000
now your salary is = 16000
-----
Process exited after 5.316 seconds with return value 0
Press any key to continue . . .

```

Write a program that read a number and print whether it is even or odd number.

```

#include<iostream>

Using namespace std;

Main()

{

Int num;

Cin>>num;

If(num%2==0)

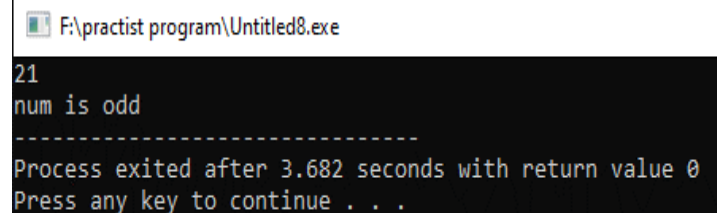
Cout<<"num is even";

Else

Cout<<"num is odd";

}

```



```

F:\practist program\Untitled8.exe
21
num is odd
-----
Process exited after 3.682 seconds with return value 0
Press any key to continue . . .

```

Write a program that read a number and prints whether it is multiple of three or not.

```

#include<iostream>

Using namespace std;

Main()

{

Int a;

Cin>>a;

If(a%3==0)

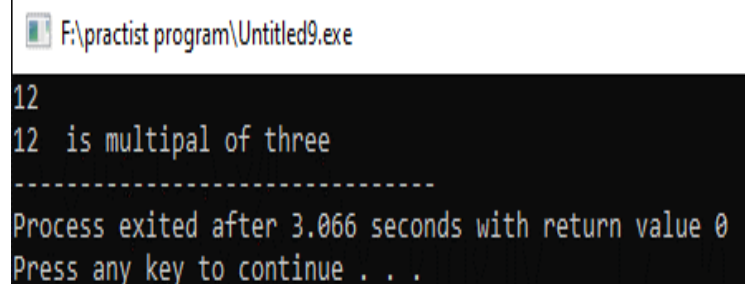
Cout<<a<<" is multiple of three";

Else

Cout<<a<<" is not multiple of three";

}

```



```

F:\practist program\Untitled9.exe
12
12 is multipal of three
-----
Process exited after 3.066 seconds with return value 0
Press any key to continue . . .

```

Write a program that reads a section of the student and print room 42 if section is A other wise room 43.

```
#include<iostream>
```

```
Using namespace std;
```

```
Main()
```

```
{
```

```
Char section;
```

```
Cin>>section;
```

```
If(section==A)
```

```
{
```

```
Cout<<" room 42";
```

```
}
```

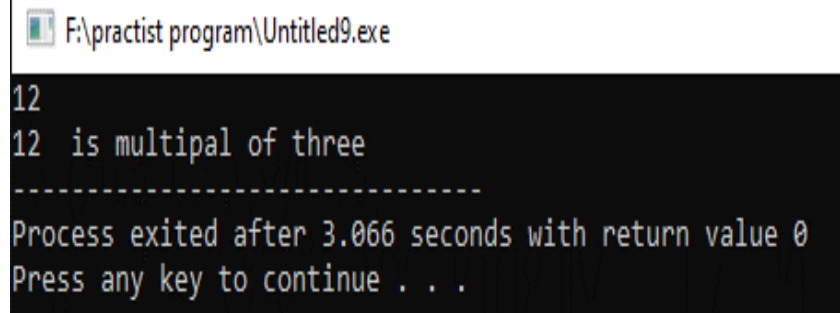
```
Else
```

```
{
```

```
Cout<<" room 43";
```

```
}
```

```
}
```



```
F:\practist program\Untitled9.exe
```

```
12
```

```
12 is multipal of three
```

```
-----
```

```
Process exited after 3.066 seconds with return value 0
```

```
Press any key to continue . . .
```

Write a program that read a number from user and print whether it is positive, negative or it is equal to zero.

```
#include<iostream>
```

```
Using namespace std;
```

```
Main()
```

```
{
```

```
Int num;
```

```
Cin>>num;
```

```
If(num>0)
```

```
{
```

```
Cout<<num<<" is positive";
```

```
}
```

```
Else if(num<0)
```

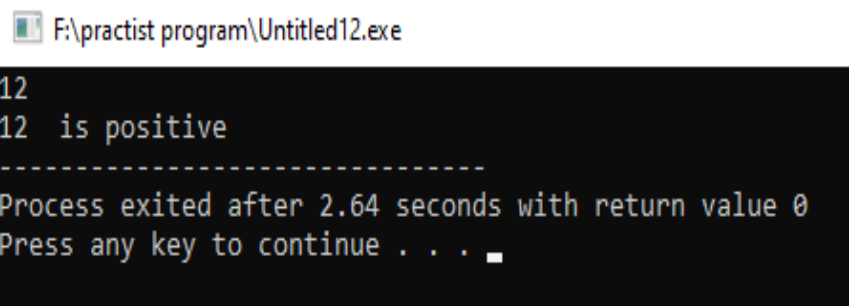
```
{
```

```
Cout<<num<<" is negative";
```

```
}
```

```
Else
```

```
{
```



```
F:\practist program\Untitled12.exe
```

```
12
```

```
12 is positive
```

```
-----
```

```
Process exited after 2.64 seconds with return value 0
```

```
Press any key to continue . . .
```

```
Cout<<num<<" is equal to zero";
```

```
}
```

```
}
```

Write a program that read the marks from user and print the grade based on the mark obtained.

```
#include<iostream>
```

```
Using namespace std;
```

```
Main()
```

```
{
```

```
Int mark;
```

```
Cin>>mark;
```

```
If(mark<=100)
```

```
{
```

```
If(mark>=80)
```

```
{
```

```
Cout<<"grade A";
```

```
}
```

```
If(mark>=80)
```

```
{
```

```
Cout<<"grade A";
```

```
}
```

```
Else If(mark>=70)
```

```
{
```

```
Cout<<"grade B";
```

```
}
```

```
Else If(mark>=60)
```

```
{
```

```
Cout<<"grade C";
```

```
}
```

```
Else If(mark>=50)
```

```
{
```

```
Cout<<"grade D";
```

```

}

Else If(mark<50)

{

Cout<<"fail";

}

}

Else

Cout<<"invalide";

}

```

```

F:\practist program\Untitled11.exe

76
grade B
-----
Process exited after 4.718 seconds with return value 0
Press any key to continue . . .

```

Write a program that read salary of the user and print output as.

If salary=50000: allowance=7%

If salary<50000: allowance=3.5%

If salary>50000: allowance=9%

```

#include<iostream>

using namespace std;

main()

{

float salary,allowance;

cout<<"enter your salary";

cin>>salary;

if(salary==50000)

{

allowance=salary*7/100;

cout<<"allowance is ="<<allowance;

cout<<"\nsalary is now"<<salary+allowance;

}

if(salary>50000)

{

allowance=salary*9/100;

cout<<"allowance is ="<<allowance;

cout<<"\nsalary is now"<<salary+allowance;

```

```
}  
  
if(salary<50000)  
{  
    allowance=salary*3.5/100;  
    cout<<"allowance is "<<allowance;  
    cout<<"\nsalary is now"<<salary+allowance;  
}  
  
}
```

F:\practist program\Untitled16.exe

enter your salary 60000

allowance is =5400

salary is now65400

Process exited after 25.93 seconds with return value 0

Press any key to continue . . . ■